

An Introduction to CNN

Exercise 1: Convolution

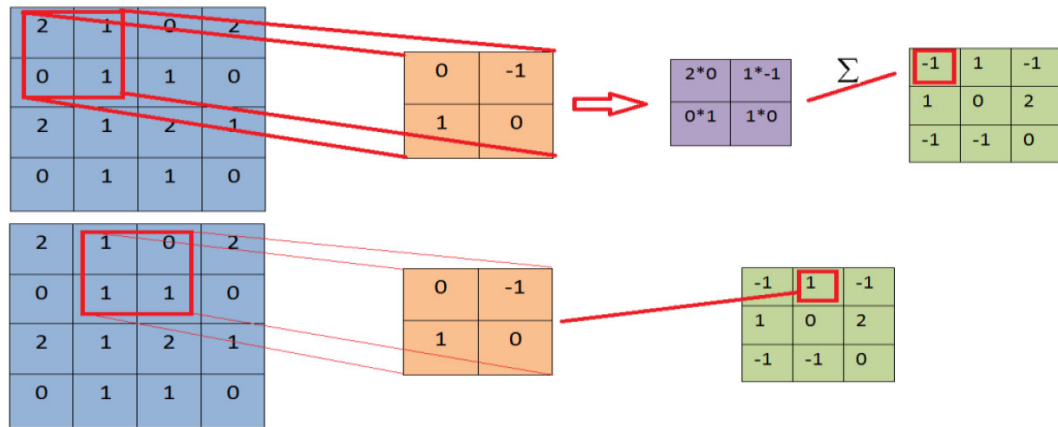


Figure 1: Demo of Convolution

In the given file directory `...\convolution_operation_exercise`, you are required to read and understand the `code.m` file, calculate the convolution results of `cat_input.jpg` using `kernel` image and visualize the results in MATLAB.

Using the six different kernels, calculate six different convolution result and check the outputs after visualization.

Exercise 2: Pooling

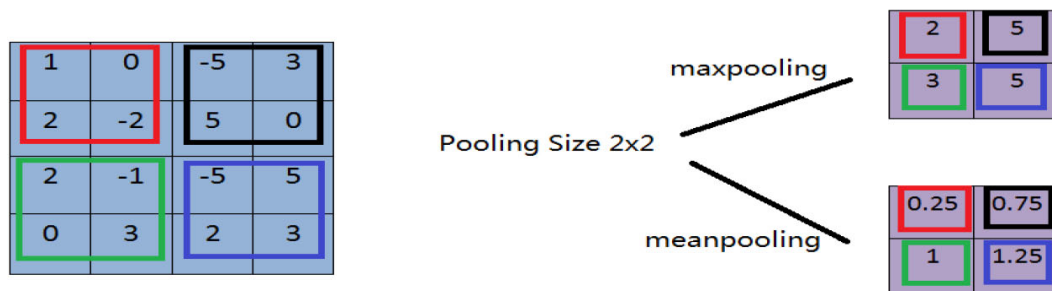


Figure 2: Demo of Pooling

In the given file directory `...\pooling_operation_exercise`, you are required to read and understand the `code.m` file, calculate the pooling results of `cat_input.jpg` using `pooling` method and visualize the results in MATLAB.

Using the two pooling methods, calculate the pooling results and check the outputs after visualization.